

EDITOR	Ramesh Chopra
EDITORIAL CORRESPONDENCE	Editorial Secretary Phone: 011-26019601 E-mail: info@efymag.com (Technical queries: efymag@efymag.com)
SUBSCRIPTIONS & MISSING ISSUES	Customer Care Officer, EFY Phone: 011-26019601 E-mail: support@efymag.com
BACK ISSUES, BOOKS, CDs, PDs, etc.	Kits'n'Games, D-48/5, Okhla Industrial Area, Phase-1 New Delhi 110033 Phone: 011-32975879, 26371661, 26371662 E-mail: kitsn@efymag.com
EXCLUSIVE NEWS/STAND DISTRIBUTOR	India Book House Pvt Ltd Arch No. 30, Below Mahatma Bridge Mahatma, Mumbai 400034 Phone: 022-4994741, 4949742, 4949743 E-mail: info@ibhworld.com
ADVERTISEMENT NEW DESK (HEAD OFFICE)	Hansa Bhatt D-47/1, Okhla Industrial Area, Phase-1, New Delhi 110020 INDIA Phone: 011-26019602, 26019603; Fax: 26017563 E-mail: info@efymag.com
MUMBAI	Flory D'Sa 207, Shivam Chambers, 48, Dr A.B. Road, World Naka Mumbai 400017 Ph: 222-345540/41, 222-345520 Fax: 22954278 E-mail: efymag@efymag.com
BANGALORE	Jayashree No. 9, 17th Main, 1st Cross, HMC II Stage Indragiri, Bangalore 560003 Ph: 080-2530594, 2530892/3; Fax: 25305994 E-mail: efymag@efymag.com
CHEMBAI	A. C. Nackeran MS House, 31-A, Cathedral Garden Road Near Palmyra Hotel, Chennai 600034 Ph: 044-3275191; Mobile: 9796250244 E-mail: efymag@efymag.com
KOLKATA	D.C. Mukherjee 61-A, Park Street, Kolkata 700016 Ph: 033-22947818; Telefax: 22650094 E-mail: efymag@efymag.com; Mobile: 09142422912
PUNE	Zakar Shah E-mail: efymag@efymag.com Mobile: 01872407753
HYDERABAD	P.S. Narasimhan Mobile: 9849942660
GULBARGA	Shambhushank Patel Ph: 079-22138840; Telefax: 079-22138841 E-mail: info@shambhushank.com; Mobile: 9882520961
SINGAPORE	My Peggy Thang, Publicity Group Pvt Ltd 72, Serendip Road, #02-20, The Laureate, Singapore 339943 Phone: +65-68522773; Fax: +65-62797302 E-mail: pthang@publicity.com, singaporemag@publicity.com
UNITED STATES	MyVernique Language, E & Tech Media Phone: +1 800 534-6677 E-mail: vernique@amamag.com
CHINA	Mr Perry Qiu, Power Pioneer Group Inc. Phone: 862126777150; Fax: 862126455239 Mobile: 136116883510, 13910897770 E-mail: tongyiguo@powerpioneer.com, qiu@powerpioneer.com
TAIWAN	Mr Jason Chen, J.K. Media 47-7, No. 348, Sec. 2, Bude Rd., Songshan Dist., Taipei City 105, Taiwan Phone: 886-2-9772780 ext. 12; Fax: 886-2-97726787

Printed, published and owned by Ramesh Chopra, Printers at Ramesh Chopra, D-48/5, Okhla Industrial Area, Phase-1, New Delhi, on the first day of each month and published from D-47/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2009. All rights reserved. Reproduction of any part of this journal from this magazine is prohibited without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility will be accepted for any error due to publishing error. Articles that cannot be used are returned to the author. If accompanied by a self-addressed and sufficiently stamped envelope, the responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrongdoings made by an advertiser. Disputes, if any, will be settled in New Delhi court only.

SUBSCRIPTION RATES					
Period (Years)	Domestic (Rs)			Overseas (US \$)	
	Print	Digital	Both	Print	Both
Five (Months)	2,360	1,445	2,880	40	—
Three (Months)	1,510	1,015	2,000	25	—
One (12 Issues)	575	385	765	10	110

For receiving Annual Directory (printed at Rs 500 with CD), please add Rs. 400 for each year (including handling and forwarding within India). Kindly add Rs 50/- for outside Delhi charges.
Please send payments only in favour of **EFY Enterprises Pvt Ltd**.
Non receipt of copies may be reported to support@efymag.com mentioning your subscription number

CONTENTS

16 | **CD-Theme**
Analogue Circuit Simulation

28 | **Technology Focus**
The Sun in A Chip: Era of Solar Devices Not Too Far

36 | **EFY Report**
The Sun is Rising in India's Solar Industry

42 | **Solar Design**
Modular Battery System For Solar Applications

49 | **Smart Cards**
Near Field Communication For Secure Applications

54 | **Futuristic**
Predicting The Future of Electronics

58 | **Construction**
H-bridge Driver Using PC

62 | **Construction**
Stereo Brick Amplifier

68 | **Career**
Robotics: Do You Have it in You?

88 | **Software Section**
Simulation of Universal Logic Board Using VHDL

100 | **Embedded Systems**
Real-time Operating Systems: It's a Dynamic World in There

105 | **Nanotechnology**
Powering Nano Devices

108 | **Special Supplement**
Certificate Programmes for Engineers

120 | **EFY Awards 2009**
Nominees for EFY Readers' Choice Awards for 2009 Announced

150 | **Interview**
"Thin Films Can Reduce Electric Power and Utility Costs"
—Dr Charles Gay, President, Applied Solar, Applied Materials

Departments

EFY-CD	08
LETTERS	12
Qs&As	14
TECHNOLOGY NEWS	20
CIRCUIT IDEAS	92
NEW PRODUCTS	116
INDUSTRY NEWS	123
BOOKSHELF	128
BIZ ADS	129
ELECTRONICS MART	141
SUBJECT & ORGANISATION INDEX	152
ADVERTISERS' INDEX	153

eStyle

- 74** | **First Look!**
- Flexi Dual Mobile Phone
 - Thin and Light Notebooks
 - Light-weight MP3 Player
 - High-quality Graphic Tablets
 - Slim Joybooks
 - Compact Laser Printers
 - Keyboard-mouse Combo
 - Slim LCD Monitors

80 | **Buyers' Guide**
Choose Your Gizmos

83 | **What's New**
3D Technology Adds New Dimension

NEXT ISSUE

TECHNOLOGY FOCUS:
Top Ten Emerging Technologies

BUYERS' GUIDE:
Smart Phones For You

SURVEY/EFY REPORT:
Forecasts for the Year